

for $y \in \{ \log(\text{reward}), \log(\text{duration}) \}$:

for $k = 1, \dots, K$:

train $\leftarrow \{ i : \text{fold}(i) \neq k \}$

Fine-tune DeBERTa backbone on $(\text{text}_i, y_i)_{i \in \text{train}}$

Discard regression head

Compute embeddings $e_i = \text{backbone}(\text{text}_i)$ for all i

Concatenate features: $X_i = [\text{hand-coded}_i, e_i]$

Fit ML learner on $(X_i, y_i)_{i \in \text{train}}$

Predict $\hat{y}_i$ on fold k (out-of-sample)